EXPERIMENT - 3

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Subject: User Interface Design
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Command line Interface(CLI)

● A Text-based Interface (Command line interface) enables users to communicate with a computer using the command line, as opposed to using a graphical interface.
● Compact and Powerful – CLI is usually faster and more powerful
than a GUI, particularly for expert users who know the correct commands.
● Automation & Scripting – The command-line interface is there to allow users to automate tasks with scripts, making the repetitive tasks easier.
● Lightweight and Resource Friendly – Compared to GUI application, CLI uses very little system resources which is why they are most suited for use on remote servers
example: Windows command prompt

IMPLEMENTATION

UID > sample.py > ...

```python
import os
import sys
def rename_file(old_name, new_name):
    try:
        os.rename(old_name,new_name)
        print(f"File renamed from {old_name} to {new_name}")
    except FileNotFoundError:
         print(f"Error: {old_name} not found.")
    except Exception as e:
        print(f"An error occured: {e}")
if __name__ == "__main__":
    if len(sys.argv) != 3:
        print(f"Usage: python rename_file_cli.py <old_filename> <new_filename>")
    else:
        rename_file(sys.argv[1], sys.argv[2])
```

RESULT

```
PS C:\Users\sruth\OneDrive\Desktop\InternEZY> & C:/Users/sruth/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/sruth/OneDrive/Desktop/InternEZY/UID/sample.py
Usage: python rename_file_cli.py <old_filename> <new_filename>
PS C:\Users\sruth\OneDrive\Desktop\InternEZY>  python sample.py sample3.py sample4.py
C:\Users\sruth\AppData\Local\Microsoft\WindowsApps\python.exe: can't open file 'C:\\Users\\sruth\\OneDrive\\Desktop\\InternEZY\\sample.py': [Errno 2] No such file or directory
PS C:\Users\sruth\OneDrive\Desktop\InternEZY> cd UID
PS C:\Users\sruth\OneDrive\Desktop\InternEZY\UID>  python sample.py sample3.py sample4.py
File renamed from sample3.py to sample4.py
PS C:\Users\sruth\OneDrive\Desktop\InternEZY\UID>
```

Graphical user interface(GUI)

- A GUI uses visual elements including icons, buttons, and windows which makes it more user-friendly for users to work with the system.
- More Resource Intensive – GUI applications use more system resources (CPU, RAM, and GPU) in contrast to a Command Line Interface (CLI) that can degrade performance in low-end devices.

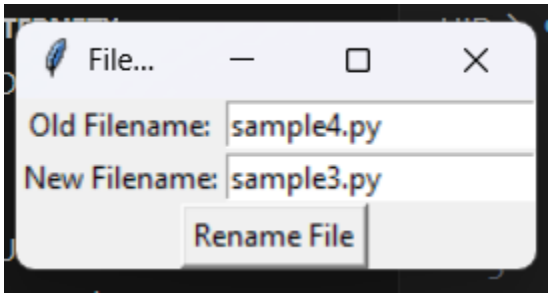
● Limited Automation: GUI does not have automation and scripting
abilities like the CLI and is thus more tedious for repetitive jobs.
example: Windows, Android, iOS

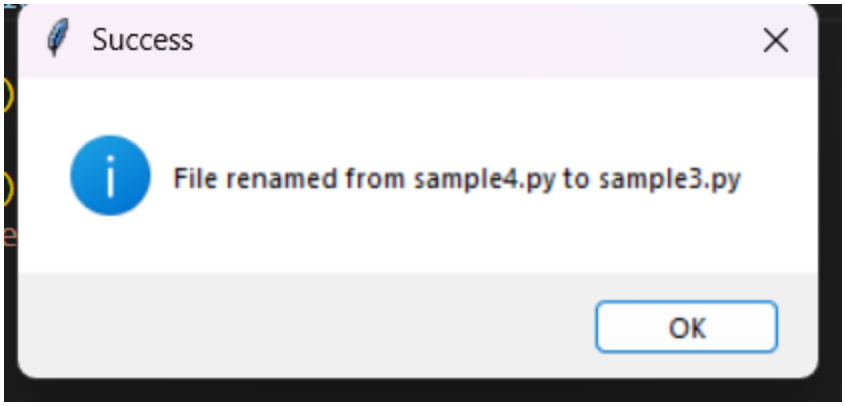
IMPLEMENTATION

UID > sample1.py > rename_file

```python
import tkinter as tk
from tkinter import messagebox
import os
def rename_file():
    old_name = old_filename_entry.get()
    new_name = new_filename_entry.get()
    try:
            os.rename(old_name, new_name)
            messagebox.showinfo("Success", f"File renamed from {old_name} to {new_name}")
    except FileNotFoundError:
        messagebox.showerror("Error", f"File {old_name} not found.")
    except Exception as e:
        messagebox.showerror("Error", f"An error occurred: {e}")
root = tk.Tk()
root.title("File Renamer")
tk.Label(root, text="Old Filename:").grid(row=0, column=0)
tk.Label(root, text="New Filename:").grid(row=1, column=0)
old_filename_entry = tk.Entry(root)
old_filename_entry.grid(row=0, column=1)
new_filename_entry = tk.Entry(root)
new_filename_entry.grid(row=1, column=1)
rename_button = tk.Button(root, text="Rename File", command=rename_file)
rename_button.grid(row=2, columnspan=2)
root.mainloop()
```

Output





Voice user interface(VUI)

• No Hands – VUI allows me to talk to a system and in some cases this is much more convenient than having to type.
• Natural Communication– VUI allows for more natural and fluid
communication, as it can process users' speech and verbal commands, eliminating typing or clicking.
• Accuracy Challenges – Speech recognition can occasionally have difficulty with accents, background noise, and complex commands, sometimes resulting in misinterpretation.
example: Amazon, Alexa, Google Assistant

IMPLEMENTATION

```python
import speech_recognition as sr
import os
def rename_file_from_voice_command(command):
    try:
        words = command.split(" ")
        old_name = words[1]
        new_name = words[3]
        os.rename(old_name, new_name)
        print(f"File renamed from {old_name} to {new_name}")
    except Exception as e:
        print(f"Error: {e}")
def listen_for_command():
    recognizer = sr.Recognizer()
    mic = sr.Microphone()
    print("Listening for command to rename a file...")
    with mic as source:
        recognizer.adjust_for_ambient_noise(source)
        audio = recognizer.listen(source)
    try:
        command = recognizer.recognize_google(audio)
        print(f"Command received: {command}")
        rename_file_from_voice_command(command)
    except sr.UnknownValueError:
        print("Sorry, I couldn't understand the command.")
    except sr.RequestError as e:
        print(f"Could not request results from Google Speech Recognition service; {e}")
if __name__ == "__main__":
    listen_for_command()
```

Output

```
(.venv) PS C:\Users\Sivaguru\OneDrive\Desktop\Sg\UID> python vui1.py
 Welcome to the Voice-Controlled File Renamer!
 Say the name of the file you want to rename (without .txt)
 You said: sample
 Say the new name for the file (without.txt)
 You said: hello
 File successfully renamed from sample.txt to hello.txt
```